SIGMIDI FOOT CONTROLLER

v0.1 / UNIFIED ELECTRICAL SCHEMATIC / external board pins only

Continuous wires connect every fitted terminal. Dots = junctions; arched crossings = no connection. Component placement is schematic.

DFRobot and display pins keep physical order. DFRobot: P2 left, P1 right, USB-C up. Outlines and distances are schematic.

3V3 / U1 logic supply

+5V

USB-C

U1 DFRobot DFR1075

Physical top view / USB-C up

P2:1 RST
P2:2 / 3V3 3V3
P2:3 NC
P2:4 / GND GND
P2:5 NC
P2:6 / GPIO23 23/SCK
P2:7 / GPIO22 22/MO
P2:8 / GPIO21 21/MI
P2:9 20/SCL
P2:10 / GPIO19 19/SDA
P2:11 9/D9
P2:12 / GPIO18 18/D7
P2:13 / GPIO1 1/D6
P2:14 NC
P2:15 / GPIO14 14/D3
P2:16 8/D2
P2:17 / GPIO16 16/TX
P2:18 / GPIO17 17/RX

P1:1 / VCC VIN
P1:2 / 3V3 3V3
P1:3 / GND GND
P1:4 / GPIO20 SCL
P1:5 SDA
P1:6 5/A4/LP_TX
P1:7 4/A3/LP_RX
P1:8 / GPIO3 3/A2
P1:9 / GPIO2 2/A1
P1:10 NC
P1:11 15/D13
P1:12 / GPIO6 6/D12/LP_SDA
P1:13 / GPIO7 7/D11/LP_SCL
P1:14 NC

All printed labels retained; unused pads left unwired.

J1 5.5 x 2.1mm

9V DC / center-negative

SLEEVE (+)
CENTER (-)

C1 47uF / 35V

U3 S13V10F5

Pololu #4083 / 1178565

VIN
5V OUT
GND

U2 Waveshare 24382

External P2 / GH1.25 / SPI

Pin 1 at top / pin 8 at bottom

1 BL
2 RST
3 D/C
4 CS
5 SCL / CLK
6 SDA / DIN
7 GND
8 VCC

Pins 1-8 in connector order

Pololu 2535 / optional LEDs

LED1 / B — 2535 — DIN, 5V, GND, DOUT
LED2 / A — 2535 — DIN, 5V, GND, DOUT
LED3 / 2 — 2535 — DIN, 5V, GND, DOUT
LED4 / 1 — 2535 — DIN, 5V, GND, DOUT
LED5 / C — 2535 — DIN, 5V, GND, DOUT
LED6 / 3 — 2535 — DIN, 5V, GND, DOUT
LED7 / D — 2535 — DIN, 5V, GND, DOUT
LED8 / 4 — 2535 — DIN, 5V, GND, DOUT
LED9 / E — 2535 — DIN, 5V, GND, DOUT
LED10 / 5 — 2535 — DIN, 5V, GND, DOUT (NC)

SW11 ENCODER

A
B

20 detents

Internal pull-ups

No external R/C

SW1 / click
C
SW2

R21 10k

J2 ACJS-MHDC

Dual TRS socket

TOP / pedal 1

11 RING
12 TIP
10 SLEEVE

7/8/9: NC

BOTTOM / pedal 2

5 RING
6 TIP
4 SLEEVE

1/2/3: NC
Pad 13: mechanical / NC

R20 5.1k
R30 5.1k
R31 10k

MODULAR FOOTSWITCH LADDER

R7 10k
R8 20k
C5 100nF
R9 1k
R10 1k
R11 1k
R12 1k
R13 1k
R14 1k
R15 1k
R16 1k
R17 1k
R18 1k

SW1 B — T1 / LED1
SW2 A — T2 / LED2
SW3 2 — T3 / LED3
SW4 1 — T4 / LED4
SW5 C — T5 / LED5
SW6 3 — T6 / LED6
SW7 D — T7 / LED7
SW8 4 — T8 / LED8
SW9 E — T9 / LED9
SW10 5 — T10 / LED10

COMMON GND

PASSIVE PEDALS

Pot: outer legs RING / SLEEVE; wiper TIP.
Variable resistor: TIP to SLEEVE; RING unused.
0-10k per input. Bias HIGH in resistor mode;
high impedance in pot / disabled / boot states.

ASSEMBLY

Set 4 / 6 / 8 / 10 switches; calibrate installed taps.
SW11 contact pin arrangement is functional.
U1 printed MO / MI mean MOSI / MISO.
USB-only supply does not power the external LED rail.

LED PACKAGE

Lens left, leads right, flat side down.
Top to bottom: DIN / 5V / GND / DOUT.
Only LEDs and wires; direct GPIO16 data.
Check 3.3V data acceptance on the actual LED batch.

Netlist, BOM and pin references: design/electrical/. All 44 components and 39 nets are checked against the connected wire drawing.